

Robel Yonas Tesfaye Bekele

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PROFILE

- **M.Sc. Software Engineering student** based in Gothenburg with a strong foundation in **Python**, **AI/perception**, and **data analysis**. I have hands-on experience building robust data pipelines (17M+ records) and applying PyTorch to real-time video analytics projects. I thrive in collaborative environments focused on psychological safety and continuous learning. I am eager to leverage my skills in a dynamic software engineering role, contributing to innovative, data-driven projects and building impactful technology solutions.

EDUCATION

- **University of Gothenburg** Gothenburg, Västra Götaland
M.Sc. in Software Engineering and Management Aug 2025 – Jun 2027 (Expected)
 - **Focus Areas:** Data-Intensive Systems, Requirements Engineering, **Quality Assurance**, Project Management
- **Kristianstad University** Kristianstad, Skåne
B.S. in Computer Science Aug 2022 – Jun 2025
 - **Concentrations:** Science with a major in Computer Sciences
 - **Coursework:** Database Systems (SQL), Data Structures, **Machine Learning**, Software Testing, Object-Oriented Design

WORK EXPERIENCE

- **Kristianstad University** Kristianstad, Skåne
Data Engineering Researcher (Thesis) Feb 2025 – Jun 2025
 - Engineered a robust data pipeline in **Python** to ingest, clean, and analyze **17M+ records**, simulating a modern **Data Warehouse** environment.
 - Optimized complex **SQL** queries for data transformation, ensuring high data quality for downstream predictive modeling.
 - Performed data modeling and addressed class imbalances to improve the accuracy of machine learning algorithms.
 - Presented technical findings to stakeholders, demonstrating the ability to translate data into **business value**.
- **Luday SE** Gothenburg, Västra Götaland
Software Engineer Intern (Full Stack) Jun 2024 – Sep 2024
 - Designed and implemented full-stack features using **Python** (Backend) and **TypeScript** (Frontend).
 - Modeled relational database schemas in **PostgreSQL** to support business requirements, ensuring data consistency.
 - Collaborated in a cross-functional Agile team, participating in sprint planning and daily stand-ups, fostering a culture of open communication and continuous learning.
 - Integrated **REST APIs** to fetch and display data, focusing on user experience and frontend visualization.
- **Youngstival** Kristianstad, Skåne
Mobile Software Engineer Oct 2024 – Jun 2025
 - Developed data-driven features using **Kotlin** and **Firebase**, managing real-time data synchronization.
 - Analyzed user feedback and app performance data to prioritize feature development and improve stability.
 - Demonstrated strong problem-solving skills by diagnosing production issues and refactoring legacy code.
 - Rapidly adapted to new technologies to meet evolving project requirements, ensuring consistent delivery.

PROJECTS

- **Parking Spot Detection System** Personal Project
Jun 2025 – Jul 2025
Software Developer
 - Developed a **PyTorch**-based video analytics pipeline to detect objects in real-time, applying **AI/ML** principles.
 - Processed unstructured video data to extract structured insights (availability counts), relevant for **IoT analytics**.
 - Optimized algorithm performance for real-time inference, balancing accuracy with computational efficiency.
- **HomeSync (IoT Control App)** Personal Project
Feb 2025 – May 2025
Software Developer
 - Engineered a **Kotlin**-based application to control **Arduino** hardware via **Socket.IO**.
 - Implemented **event-driven communication** to ensure fault tolerance and low-latency data flow.
 - Integrated hardware and software components, demonstrating competence with **embedded systems** and **sensor integration**.

SKILLS

- **Data & AI:** Python (Pandas/NumPy), **SQL**, **Data Modeling**, **Machine Learning**, PyTorch, Data Visualization
- **Development:** Python, TypeScript, Java, **REST APIs**, Git, Docker, CI/CD
- **Databases:** PostgreSQL, MySQL, Firebase (NoSQL), Database Design
- **Soft Skills:** Collaborative Mindset, **Curious & Open to Learning**, Problem Solving, Stakeholder Communication, Agile/Scrum
- **Languages:** Swedish (Fluent), English (Fluent), Amharic (Native)

REFERENCES

Dayo Ajakaiye (CTO, Luday AB): dayo@luday.se

Tim Omorogieva (Youngstival): tim@youngstival.se